Muhammad Ariff Bin Abdul Razak

+65 96579407 | muhdariffrazak00@gmail.com | www.linkedin.com/in/muhd-ariff-razak | Singapore

WORK EXPERIENCE

KLASS Engineering and Solutions
Machine Learning & Fullstack Engineer (Intern)
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, SpeechForensics) on SLURM-based HPC to assess detection capability against AI-generated media as a public-safety threat, achieving 90.03 AUROC on the NSG-VD replication.
- Ran ablation studies integrating optical flow (RAFT), segmentation (SAM 2), and point tracking (CoTracker) to test whether physical-motion signals in video sequences can distinguish AI-generated content, finding mixed evidence of a usable signal.
- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-sync (LatentSync) models to curate a balanced 1,200-video real/fake dataset, used to compare all four detectors consistently and to test models at small scale before committing larger compute.
- Evaluated audio-visual synchronization as a deepfake detection signal, selecting a ~0.3 operating threshold through ROC analysis.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits that caught pipeline errors before committing to training and synthetic-deepfake generation runs of up to 5 hours each.
- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for a company-wide internal AI chatbot platform that gave staff a secure alternative to public AI tools, with search across internal documents.

SUTD – ROAR Lab
Research Assistant
Aug 2022 – May 2024 (Volunteer until Nov 2022, Full Time thereafter)

- Led the design of Panthera's intelligent lighting control system to signal the autonomous vehicle's intent to pedestrians and meet testing safety requirements, running a rosserial node on an Arduino under an RTOS that mapped cmd_vel, IMU, and vehicle-state topics to turn, warning, and hazard signals; the platform passed T1 autonomous testing at CETRAN.
- Wired and integrated the 3D LiDAR used for Panthera's obstacle detection, mapping, and localization, verifying sensor output through depth-map visualization, and supported the vehicle through its T1 autonomous field testing.
- Explored predictive safety for a stair-climbing robot by training and evaluating Long Short-Term Memory (LSTM) networks on its sensor time-series data, showing that fall risk could be predicted.

Singapore Institute of Technology (SIT)
Student Research Engineer
Aug 2024 – May 2025

- Designed and built a temperature-controlled enclosure around a compression testing machine for faculty research on how foods behave during chewing, defining the electrical architecture and selecting components before handing the completed chamber over to the research team.
- Engineered closed-loop temperature control with a thermistor and standalone temperature controller, holding the chamber within ±2 °C of a 37 °C body-temperature setpoint for consistent test conditions.

EDUCATION

DigiPen Institute of Technology & SIT
Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic
Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic
Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Relevant Coursework: Microcontroller Applications, PLCs, and Robotics Integration

PROJECTS

1. AI-Generated Video & Audio Forensics Benchmarking
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, SpeechForensics) on SLURM-based HPC to assess detection capability against AI-generated media as a public-safety threat, achieving 90.03 AUROC on the NSG-VD replication.
- Ran ablation studies integrating optical flow (RAFT), segmentation (SAM 2), and point tracking (CoTracker) to test whether physical-motion signals in video sequences can distinguish AI-generated content, finding mixed evidence of a usable signal.
- Evaluated audio-visual synchronization as a deepfake detection signal, selecting a ~0.3 operating threshold through ROC analysis.

2. Synthetic Multimodal Deepfake Dataset Pipeline
Sep 2025 – Aug 2026

- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-sync (LatentSync) models to curate a balanced 1,200-video real/fake dataset, used to compare all four detectors consistently and to test models at small scale before committing larger compute.

3. Multi-Model AI Endpoint Platform
Sep 2025 – Aug 2026

- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for a company-wide internal AI chatbot platform that gave staff a secure alternative to public AI tools, with search across internal documents.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits that caught pipeline errors before committing to training and synthetic-deepfake generation runs of up to 5 hours each.

4. Stair-Climbing Robot Fall-Risk Prediction (LSTM)
Aug 2022 – May 2024

- Explored predictive safety for a stair-climbing robot by training and evaluating Long Short-Term Memory (LSTM) networks on its sensor time-series data, showing that fall risk could be predicted.

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics: Deep Learning, Computer Vision, Multimodal ML, Deepfake Detection, Generative AI, OpenCV, Optical Flow (RAFT), Segmentation (SAM 2), Point-Tracking (CoTracker)
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNNs, DNNs, LSTMs, YOLO, Vision Transformers, Reinforcement Learning, Model Benchmarking, Zero-shot Learning
Engineering Data & Analysis: Applied Machine Learning, Predictive Modeling, Regression Analysis, Classification Models, Scikit-learn, Data Preprocessing, Model Evaluation & Validation, Time Series Analysis, Sensor Data Analysis, Fault Detection, Data Visualization, Data-driven Engineering Optimization, Industrial Automation (PLCs), Engineering Data Analysis & Interpretation, Data-driven Decision Making
Software & Fullstack: Python (NumPy, Pandas), C++17, C#, Java, TypeScript, React, Zustand, FastAPI, OpenAPI, WebSockets, Git, Linux/Ubuntu, WSL
Infrastructure & Cloud Tech: SLURM, GPU Computing (CUDA), NVIDIA HPC Clusters (H200 nodes), Docker, Dev Containers, ONNX Runtime, uv, IoT Integrations, Agile/Scrum Methodologies

AWARDS / ACHIEVEMENTS / CERTIFICATES

- MOE (Ministry of Education) Instructor Certification (Achieved at age 18) — Valid Oct 2018 – Oct 2026 (renewing)
- Google Crash Course on Python
- Google The Bits and Bytes of Computer Networking
- Coursera Machine Learning